

Lab 1: Film Ranking (The Basics)

```
1  WITH ranked_films AS (  
2      SELECT  
3          c.name AS category_name,  
4          f.title,  
5          f.length,  
6          DENSE_RANK() OVER (  
7              PARTITION BY c.name  
8              ORDER BY f.length DESC  
9          ) AS length_rank  
10     FROM film f  
11     JOIN film_category fc  
12         ON f.film_id = fc.film_id  
13     JOIN category c  
14         ON fc.category_id = c.category_id  
15 )  
16 SELECT  
17     category_name,  
18     title,  
19     length,  
20     length_rank  
21 FROM ranked_films  
22 WHERE length_rank <= 3  
23 ORDER BY  
24     category_name,  
25     length_rank,  
26     title;
```

	category_name character varying (25) 🔒	title character varying (255) 🔒	length smallint 🔒	length_rank bigint 🔒
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	1
3	Action	Casualties Encino	179	2
4	Action	Quest Mussolini	177	3
5	Animation	Gangs Pride	185	1
6	Animation	Pond Seattle	185	1
7	Animation	Sons Interview	184	2
8	Animation	Theory Mermaid	184	2
9	Animation	Lawless Vision	181	3
10	Children	Fury Murder	178	1
11	Children	Wrong Behavior	178	1
12	Children	Empire Malkovich	177	2
13	Children	Model Fish	175	3
14	Classics	Conspiracy Spirit	184	1
15	Classics	Dracula Crystal	176	2
16	Classics	Lights Deer	174	3
17	Comedy	Control Anthem	185	1
18	Comedy	Saturn Name	182	2
19	Comedy	Searchers Wait	182	2
20	Comedy	Caper Motions	176	3
21	Comedy	Sweden Shining	176	3

22	Documentary	Wife Turn	183	1
23	Documentary	Young Language	183	1
24	Documentary	Cause Date	179	2
25	Documentary	Dozen Lion	177	3
26	Drama	Jacket Frisco	181	1
27	Drama	Something Duck	180	2
28	Drama	Slacker Liaisons	179	3
29	Drama	Torque Bound	179	3
30	Drama	Virgin Daisy	179	3
31	Family	King Evolution	184	1
32	Family	Baked Cleopatra	182	2
33	Family	Manchurian Curtain	177	3
34	Foreign	Crystal Breaking	184	1
35	Foreign	Sorority Queen	184	1
36	Foreign	Catch Amistad	183	2
37	Foreign	Hotel Happiness	181	3
38	Foreign	Intrigue Worst	181	3
39	Games	Chicago North	185	1
40	Games	Moonwalker Fool	184	2
41	Games	Monsoon Cause	182	3

42	Horror	Analyze Hoosiers	181	1
43	Horror	Love Suicides	181	1
44	Horror	Texas Watch	179	2
45	Horror	Patton Interview	175	3
46	Music	Home Pity	185	1
47	Music	Scalawag Duck	183	2
48	Music	Reds Pocus	182	3
49	New	Frontier Cabin	183	1
50	New	Runaway Tenenbaums	181	2
51	New	Wild Apollo	181	2
52	New	Redemption Comforts	179	3
53	Sci-Fi	Soldiers Evolution	185	1
54	Sci-Fi	Express Lonely	178	2
55	Sci-Fi	Connecticut Tramp	172	3
56	Sci-Fi	Eyes Driving	172	3
57	Sports	Smoochy Control	184	1
58	Sports	Records Zorro	182	2
59	Sports	Star Operation	181	3
60	Travel	Muscle Bright	185	1
61	Travel	Sweet Brotherhood	185	1
62	Travel	Born Spinal	179	2
63	Travel	Kick Savannah	179	2
64	Travel	Italian African	174	3

Lab 2: Growth Analysis (Time Series)

```
29 WITH monthly_revenue AS (  
30     SELECT  
31         DATE_TRUNC('month', payment_date) AS month,  
32         SUM(amount) AS revenue  
33     FROM payment  
34     GROUP BY DATE_TRUNC('month', payment_date)  
35 ),  
36 revenue_with_previous AS (  
37     SELECT  
38         month,  
39         revenue,  
40         LAG(revenue, 1) OVER (  
41             ORDER BY month  
42         ) AS previous_revenue  
43     FROM monthly_revenue  
44 )  
45 SELECT  
46     month,  
47     revenue,  
48     previous_revenue,  
49     ROUND(  
50         ((revenue - previous_revenue) / previous_revenue) * 100,  
51         2  
52     ) AS growth_percent  
53 FROM revenue_with_previous  
54 ORDER BY month;
```

	month timestamp without time zone 🔒	revenue numeric 🔒	previous_revenue numeric 🔒	growth_percent numeric 🔒
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value (Running Total)

```
59  --lab3--
60  SELECT
61      customer_id,
62      payment_id,
63      payment_date,
64      amount,
65      SUM(amount) OVER (
66          PARTITION BY customer_id
67          ORDER BY payment_date
68          ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
69      ) AS cumulative_spending
70  FROM payment
71  ORDER BY
72      customer_id,
73      payment_date;
```

Data Output Messages Notifications

	customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_spending numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92
9	1	22681	2007-03-02 14:05:18.996577	3.99	40.91
10	1	22682	2007-03-02 16:30:04.996577	0.99	41.90
11	1	22683	2007-03-17 11:06:20.996577	4.99	46.89

Lab 4: The "Top-N" Filter Challenge

```

75  --lab4--
76  WITH customer_spending AS (
77      SELECT
78          co.country,
79          c.customer_id,
80          c.first_name,
81          c.last_name,
82          SUM(p.amount) AS total_spent
83      FROM customer c
84      JOIN address a ON c.address_id = a.address_id
85      JOIN city ci ON a.city_id = ci.city_id
86      JOIN country co ON ci.country_id = co.country_id
87      JOIN payment p ON c.customer_id = p.customer_id
88      GROUP BY
89          co.country,
90          c.customer_id,
91          c.first_name,
92          c.last_name),
93  ranked_customers AS (
94      SELECT
95          country,
96          customer_id,
97          first_name,
98          last_name,
99          total_spent,
100     RANK() OVER (
101         PARTITION BY country
102         ORDER BY total_spent DESC
103     ) AS spending_rank
104     FROM customer_spending)
105  SELECT
106     country,
107     customer_id,
108     first_name,
109     last_name,
110     total_spent,
111     spending_rank
112  FROM ranked_customers
113  WHERE spending_rank = 1
114  ORDER BY country;

```

	country character varying (50) 🔒	customer_id integer 🔒	first_name character varying (45) 🔒	last_name character varying (45) 🔒	total_spent numeric 🔒	spending_rank bigint 🔒
1	Afghanistan	218	Vera	Mccoy	67.82	1
2	Algeria	176	June	Carroll	151.68	1
3	American Samoa	320	Anthony	Schwab	47.85	1
4	Angola	528	Claude	Herzog	93.80	1
5	Anguilla	381	Bobby	Boudreau	99.68	1
6	Argentina	560	Jordan	Archuleta	129.71	1
7	Armenia	41	Stephanie	Mitchell	118.75	1
8	Austria	266	Nora	Herrera	113.74	1
9	Azerbaijan	334	Raymond	Mcwhorter	108.75	1
10	Bahrain	590	Seth	Hannon	108.76	1
11	Bangladesh	21	Michelle	Clark	146.68	1
12	Belarus	144	Clara	Shaw	189.60	1
13	Bolivia	431	Joel	Francisco	92.78	1
14	Brazil	178	Marion	Snyder	194.61	1
15	Brunei	91	Lois	Butler	107.66	1
16	Bulgaria	540	Tyrone	Asher	112.76	1
17	Cambodia	516	Elmer	Noe	92.76	1